

Significance (or hypothesis) tests

Quantiles

Def. Let f be a density function and $n \in \mathbb{N}$. The values $\xi_1, \dots, \xi_{n-1}$ determined such that

$$\int_{-\infty}^{\xi_1} f(x)dx = \int_{\xi_1}^{\xi_2} f(x)dx = \dots = \int_{\xi_{n-1}}^{\infty} f(x)dx = \frac{1}{n} \stackrel{\text{not}}{=} \alpha$$

are called quantiles or α -quantiles.

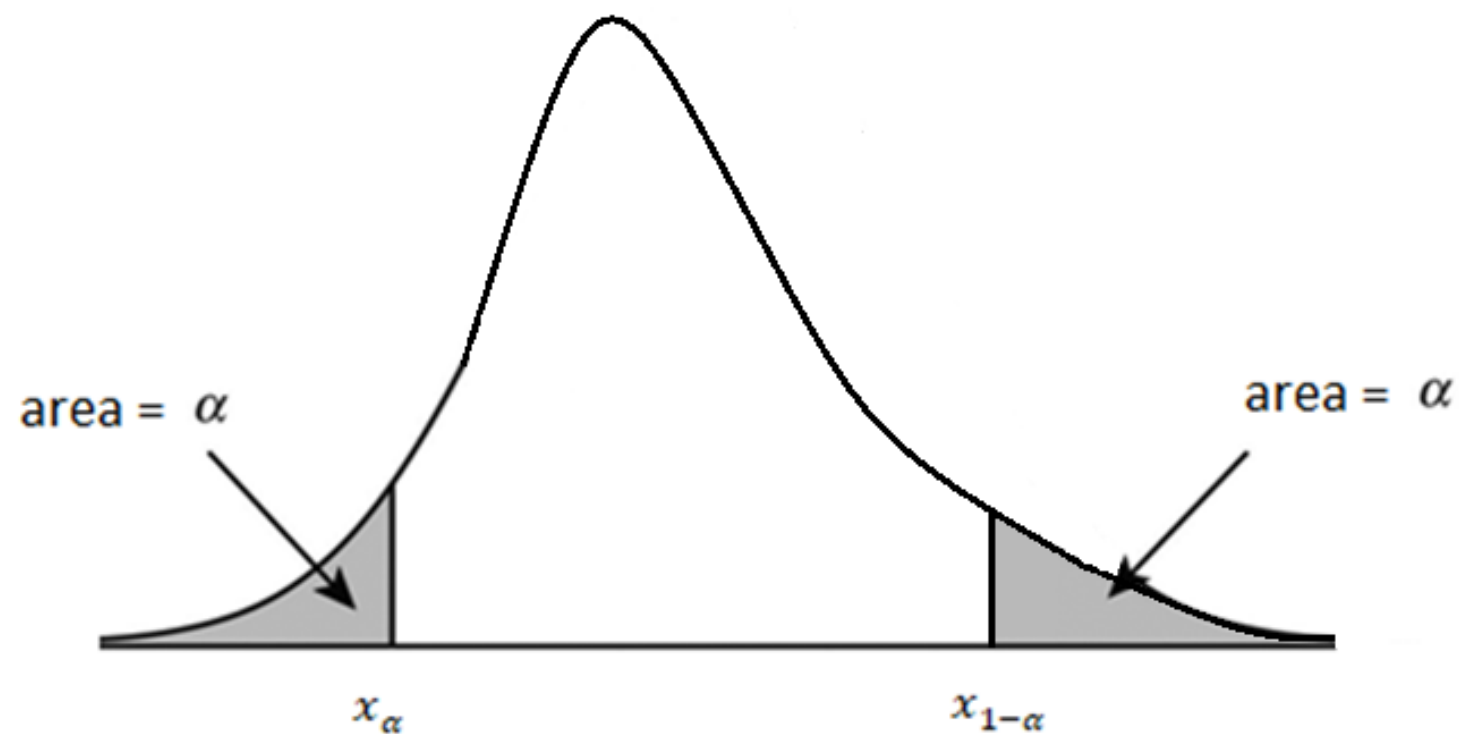
In the special cases where $\alpha = \frac{1}{2}, \frac{1}{4}, \frac{1}{10}, \frac{1}{100}$, these α -quantiles are called median, quartiles, deciles, percentiles.

Quantiles

ξ_1 is the inferior α -quantile $\stackrel{\text{not}}{=} x_\alpha$

ξ_{n-1} is the superior α -quantile $= x_{1-\alpha}$

$F(\xi_1) = 1 - F(\xi_{n-1}) = \alpha$, where F is the distribution function



χ^2 , t and F distributions

1) χ^2 (chi-squared) with k degrees of freedom

$$\chi_k^2 = \sum_{i=1}^k U_i^2, \text{ where } U_i \sim i.i.d. N(0,1), i = \overline{1, k}$$

The distribution function is

$$H_k(x) = \frac{1}{2^{\frac{k}{2}} \cdot \Gamma\left(\frac{k}{2}\right)} \int_0^x u^{\frac{k}{2}-1} \cdot e^{-\frac{u}{2}} du,$$

where $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} du$.

χ^2 , t and F distributions

2) t (or Student's) distribution with k degrees of freedom

$$t_k = \frac{U}{\sqrt{\frac{\chi_k^2}{k}}}, \quad U \sim N(0,1) \text{ independent of } \chi_k^2.$$

The distribution function is

$$G_k(z) = \frac{1}{\sqrt{\pi k}} \frac{\Gamma\left(\frac{k+1}{2}\right)}{\Gamma\left(\frac{k}{2}\right)} \int_{-\infty}^z \left(1 + \frac{x^2}{k}\right)^{-\frac{k+1}{2}} dx.$$

Its α -quantiles satisfy the relation

$$t_{k,\alpha} = -t_{k,1-\alpha}.$$

χ^2 , t and F distributions

3) F (or Snedecor's) distribution

$$F_{k_1, k_2} = \frac{k_2 \cdot \chi_{k_1}^2}{k_1 \cdot \chi_{k_2}^2}, \quad \chi_{k_1}^2 \text{ and } \chi_{k_2}^2 \text{ are independent.}$$

The distribution function is

$$S_{k_1, k_2} = \left(\frac{k_1}{k_2}\right)^{\frac{k_1}{2}} \frac{\Gamma\left(\frac{k_1+k_2}{2}\right)}{\Gamma\left(\frac{k_1}{2}\right)\Gamma\left(\frac{k_2}{2}\right)} \int_0^x z^{\frac{k_1}{2}-1} \cdot \left(1 + \frac{k_1}{k_2} \cdot z\right)^{-\frac{k_1+k_2}{2}} dz.$$

Significance tests

They are statistical tools to measure whether a set of observed data produces a result more extreme than what chance might produce. If the result is outside the limits of chance variation, it is said to be *statistically significant*.

Suppose that $\{X_1, \dots, X_n\}$ are *i.i.d.* as the random variable X .

We are interested to check whether X satisfies the hypothesis H_0 (called *null hypothesis*) against H_A (called *alternative hypothesis*).

A test is a criterion (a rule) based upon $X_1, \dots, X_n$.

Significance tests

A classical procedure to build a test is as follows:

- the sample space S^n is divided in $W \cup \bar{W} = S^n$
- if the statistical data $(x_1, \dots, x_n)' \in \bar{W}$ then H_0 is rejected;
if $(x_1, \dots, x_n)' \in W$ then H_0 is accepted (or *fail to reject*).
- $\bar{W}$ is called the critical region of the test and it is determined such that $P((X_1, \dots, X_n)' \in \bar{W} \mid H_0 \text{ is true}) = \alpha$, where α is a given value (usually 0.05 or 0.01) called *significance level*.
- W is called the acceptance region of the test.

Significance tests

Decision \ True hypothesis	H_0	H_A
H_0	$(H_0 H_0)$	$(H_0 H_A)$
H_A	$(H_A H_0)$	$(H_A H_A)$

Type II error = failure to reject a false null hypothesis

Type I error = rejection of a true null hypothesis

$\alpha = P(H_A|H_0)$ – risk of type I

$\beta = P(H_0|H_A)$ – risk of type II

$\pi = 1 - \beta = P(H_A|H_A)$ – power of the test

We are interested in tests with α, β small, π big.

Confidence intervals

Let X be a random variable with the distribution F_θ , $\theta \in \Theta \subseteq \mathbb{R}$ and $X_1, \dots, X_n$ a random sample *i.i.d.* as X .

If $\alpha \in (0,1)$ (small) and $A_\alpha, B_\alpha : S^n \rightarrow \mathbb{R}$ are two measurable functions such that:

$$\begin{aligned} i) & A_\alpha(x_1, \dots, x_n) \leq B_\alpha(x_1, \dots, x_n), \quad \forall (x_1, \dots, x_n)' \in S^n \\ ii) & P_\theta(A_\alpha(X_1, \dots, X_n) \leq \theta \leq B_\alpha(X_1, \dots, X_n)) = 1 - \alpha. \end{aligned}$$

then, for the statistical data $(x_1, \dots, x_n)$, the interval

$$C_{n;1-\alpha}(x_1, \dots, x_n) = [A_\alpha(x_1, \dots, x_n), B_\alpha(x_1, \dots, x_n)]$$

is called *confidence interval* for θ , with the *confidence level* $1 - \alpha$.

Confidence intervals

How to determine confidence intervals – the method of the pivotal function

Def. $g : S^n \times \Theta \rightarrow \mathbb{R}$ is called the pivotal function *iff*:

- $g((x_1, \dots, x_n), \theta)$ is continuous and strictly monotone with regard to θ ;
- $g((X_1, \dots, X_n), \theta)$ has the distribution independent of θ .

Given $\alpha \in (0,1)$, we determine two quantiles $a(\alpha), b(\alpha)$ of the distribution of g , such that

$$P_{\theta} \left(a(\alpha) \leq g((X_1, \dots, X_n), \theta) \leq b(\alpha) \right) = 1 - \alpha.$$

Confidence intervals

The confidence interval $C_{n;1-\alpha}(x_1, \dots, x_n) = [A_\alpha(x_1, \dots, x_n), B_\alpha(x_1, \dots, x_n)]$ is determined by solving the two inequations:

$$a(\alpha) \leq g((x_1, \dots, x_n), \theta)$$

$$g((x_1, \dots, x_n), \theta) \leq b(\alpha).$$

Obs. The quantiles $a(\alpha)$ and $b(\alpha)$ are not uniquely determined, therefore the confidence interval is not unique. We are interested to determine the shortest confidence interval, given a confidence level.

Confidence intervals

Example $X \sim N(\mu, \sigma^2)$, μ unknown; σ^2 known

$X_1, \dots, X_n$ a random sample for X

$$\left[\begin{array}{l} Y_i \sim N(\mu_i, \sigma_i^2) \text{ indep.} \\ Y = a_1 Y_1 + \dots + a_n Y_n \end{array} \right. \Rightarrow Y \sim N\left(\sum_{i=1}^n a_i \mu_i, \sum_{i=1}^n a_i^2 \mu_i^2\right)$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

We take the pivotal function

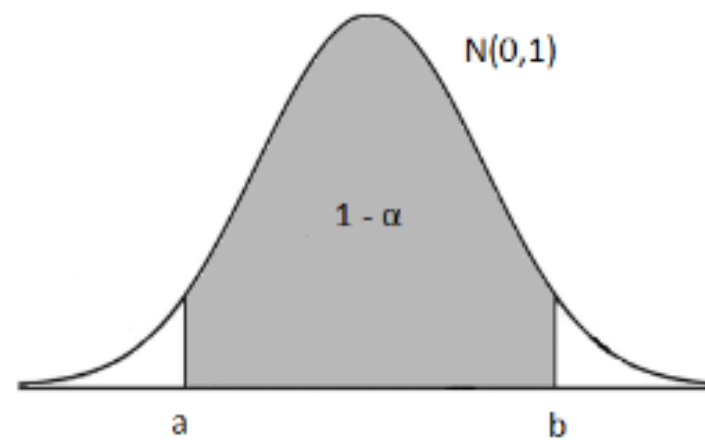
$$g((X_1, \dots, X_n); \mu) = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1) - \text{does not depend on } \mu$$

and g is strictly decreasing with regard to μ

Confidence intervals

For a given $\alpha \in (0,1)$, let a, b be two quantiles of $N(0,1)$ such that

$$P\left(a \leq \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq b\right) = 1 - \alpha$$



(a,b) can be asymmetric

$x_1, \dots, x_n$ – statistical data

$$a \leq \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq b \Leftrightarrow \bar{x} - b \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} - a \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow C_{n;1-\alpha} = \left[\bar{x} - b \frac{\sigma}{\sqrt{n}}, \bar{x} - a \frac{\sigma}{\sqrt{n}} \right].$$

Confidence intervals

We are interested in finding the shortest interval

$$l = \frac{\sigma}{\sqrt{n}}(b - a) \quad \text{the length of } C_{n;1-\alpha}$$

$$b \text{ depends on } a: b = b(a)$$

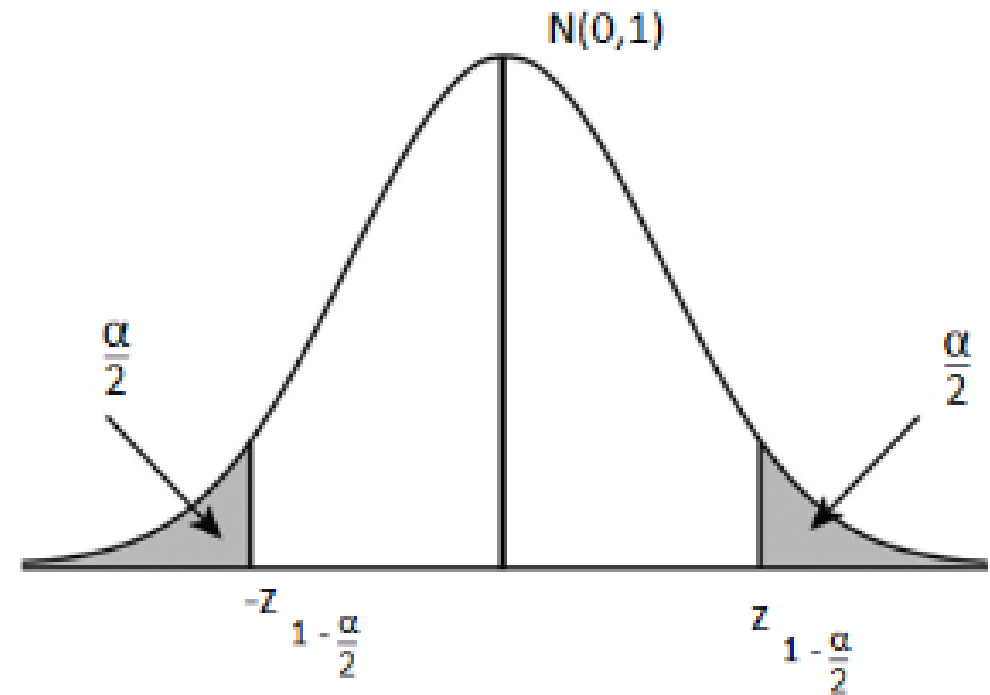
$$\begin{cases} F_{N(0,1)}(b) - F_{N(0,1)}(a) = 1 - \alpha, & F_{N(0,1)} = \text{distribution function of } N(0,1) \\ \min\{\frac{\sigma}{\sqrt{n}}(b - a)\} \end{cases}$$

$$\begin{cases} f_{N(0,1)}(b) \cdot \frac{\partial b}{\partial a} - f_{N(0,1)}(a) = 0 \\ \frac{\partial b}{\partial a} - 1 = 0 \end{cases}$$

Confidence intervals

$\Rightarrow f_{N(0,1)}(b) = f_{N(0,1)}(a) \Rightarrow$ the shortest confidence interval is the symmetric one

$$b = z_{1-\frac{\alpha}{2}} \quad a = -z_{1-\frac{\alpha}{2}}$$



$\Rightarrow C_{n;1-\alpha}^*(x_1, \dots, x_n) = [\bar{x} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}]$ is the shortest confidence interval for μ , with the confidence level $1 - \alpha$.

If $\alpha = 0.05 \Rightarrow z_{1-\frac{\alpha}{2}} = 1.96$.

Significance tests

How to build a significance test – the intuitive method based upon acceptance regions

$H_0: \{\theta = \theta_0\}$ simple null hypothesis

$H_A: \{\theta \neq \theta_0\}$ composite alternative hypothesis

(can have many different θ_s)

α is the significance level (given).

Suppose that there is a pivotal function g , then we have two quantiles $a(\alpha)$ and $b(\alpha)$ such that:

$$P_{\theta_0}(a(\alpha) \leq g((X_1, \dots, X_n); \theta_0) \leq b(\alpha)) = 1 - \alpha.$$

Significance tests

We set the acceptance region of H_0 at the significance level α as follows:

$$W_{n;1-\alpha}(\theta_0) = \{(x_1, \dots, x_n)' \mid a(\alpha) \leq g((x_1, \dots, x_n); \theta_0) \leq b(\alpha)\}$$

and the critical region as complement $\bar{W}_{n;1-\alpha}(\theta_0)$.

The probability of type I error is

$$P(H_A|H_0) = P_{\theta_0} \left((X_1, \dots, X_n) \in \bar{W}_{n;1-\alpha}(\theta_0) \right) = 1 - (1 - \alpha) = \alpha.$$

Significance tests

Most applications of hypothesis testing control the probability of making a type I error, but it is not always the case with the type II error. With this regard, an operating characteristic curve can be plotted to show how the sample size affects the probability of making a type II error.

The operating characteristic function of the test is defined as follows:

$$OC(\theta) = P_{\theta} \left((X_1, \dots, X_n) \in W_{n;1-\alpha}(\theta_0) \right).$$

Usually, the smaller the difference between θ and θ_0 , the larger the sample size n needs to be in order to get a small β .

Significance tests

For example, if the true θ is 1.5 and we test $H_0: \{\theta = 1.7\}$, the probability of accepting H_0 when H_A is true is very high. If we increase the sample size n , the probability for the test to perceive the difference (i.e., H_0 is rejected) increases.

On the other hand, if we test $H_0: \{\theta = 2.5\}$, we expect a much lower probability of accepting H_0 .

One sample t -test

It applies for the mean of a normal distribution with unknown variance.

$$X \sim N(\mu, \sigma^2), \quad \sigma^2 \text{ unknown}$$

$$X_1, \dots, X_n \text{ a random sample } N(\mu, \sigma^2)$$

$$H_0: \{\mu = \mu_0\} \quad H_A: \{\mu \neq \mu_0\}.$$

One sample t -test

I. First, we determine the confidence interval for μ , with the confidence level $1 - \alpha$.

Recall that $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is MLE for μ – it is unbiased

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ is unbiased for } \sigma^2.$$

It can be proven that $\frac{n-1}{\sigma^2} S^2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 \sim \chi^2(n-1)$ and $\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}$ is independent from $\frac{n-1}{\sigma^2} S^2$ (for proof, see Dumitrescu & Bățătorescu, pag 176).

It follows that $\frac{\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}}{\sqrt{\frac{n-1}{\sigma^2} S^2 \frac{1}{n-1}}} = \frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}$.

One sample t -test

Now, we take the pivotal function $g((x_1, \dots, x_n); \mu) = \frac{\sqrt{n}(\bar{x} - \mu)}{s}$, g is strictly decreasing in μ and its distribution t_{n-1} does not depend on μ .

For $\alpha \in (0,1)$, there are two quantiles $a(\alpha)$, $b(\alpha)$ for the distribution t_{n-1} such that

$$P_{\mu} \left(a(\alpha) \leq \frac{\sqrt{n}(\bar{X} - \mu)}{s} \leq b(\alpha) \right) = 1 - \alpha.$$

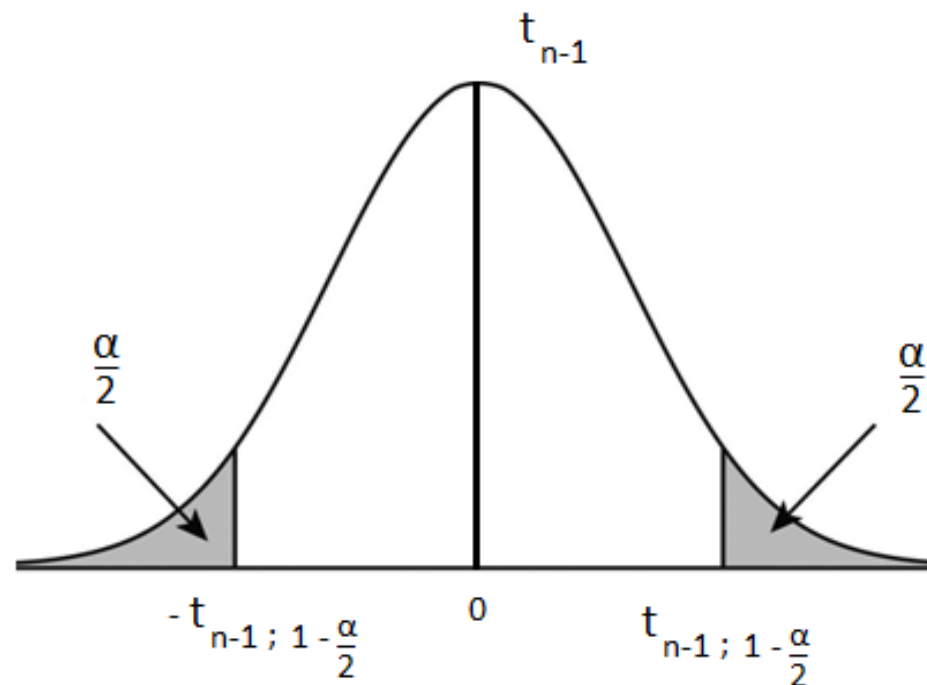
One sample t -test

Thus, $C_{n;1-\alpha} = [\bar{x} - b \frac{s}{\sqrt{n}}, \bar{x} - a \frac{s}{\sqrt{n}}]$ is a confidence interval for μ , with the confidence level $1 - \alpha$.

The shortest confidence interval is the symmetric one, where

$$b = t_{n-1;1-\frac{\alpha}{2}} \quad a = -t_{n-1;1-\frac{\alpha}{2}}$$

(the proof is similar to the example at pages 14-15)



One sample t -test

II. Now, we build the one-sample t -test

The acceptance region for $H_0: \{\mu = \mu_0\}$ is the following:

$$\begin{aligned} W_{n;1-\alpha}(\mu_0) &= \{(x_1, \dots, x_n) \mid -t_{n-1;1-\frac{\alpha}{2}} \leq \frac{\sqrt{n}(\bar{x}-\mu_0)}{s} \leq t_{n-1;1-\frac{\alpha}{2}}\} \\ &= \left\{ (x_1, \dots, x_n) \mid \mu_0 - t_{n-1;1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \leq \bar{x} \leq \mu_0 + t_{n-1;1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right\}. \end{aligned}$$

The critical region is the complement $\bar{W}_{n;1-\alpha}(\mu_0)$.

One sample t -test

The operating characteristic function

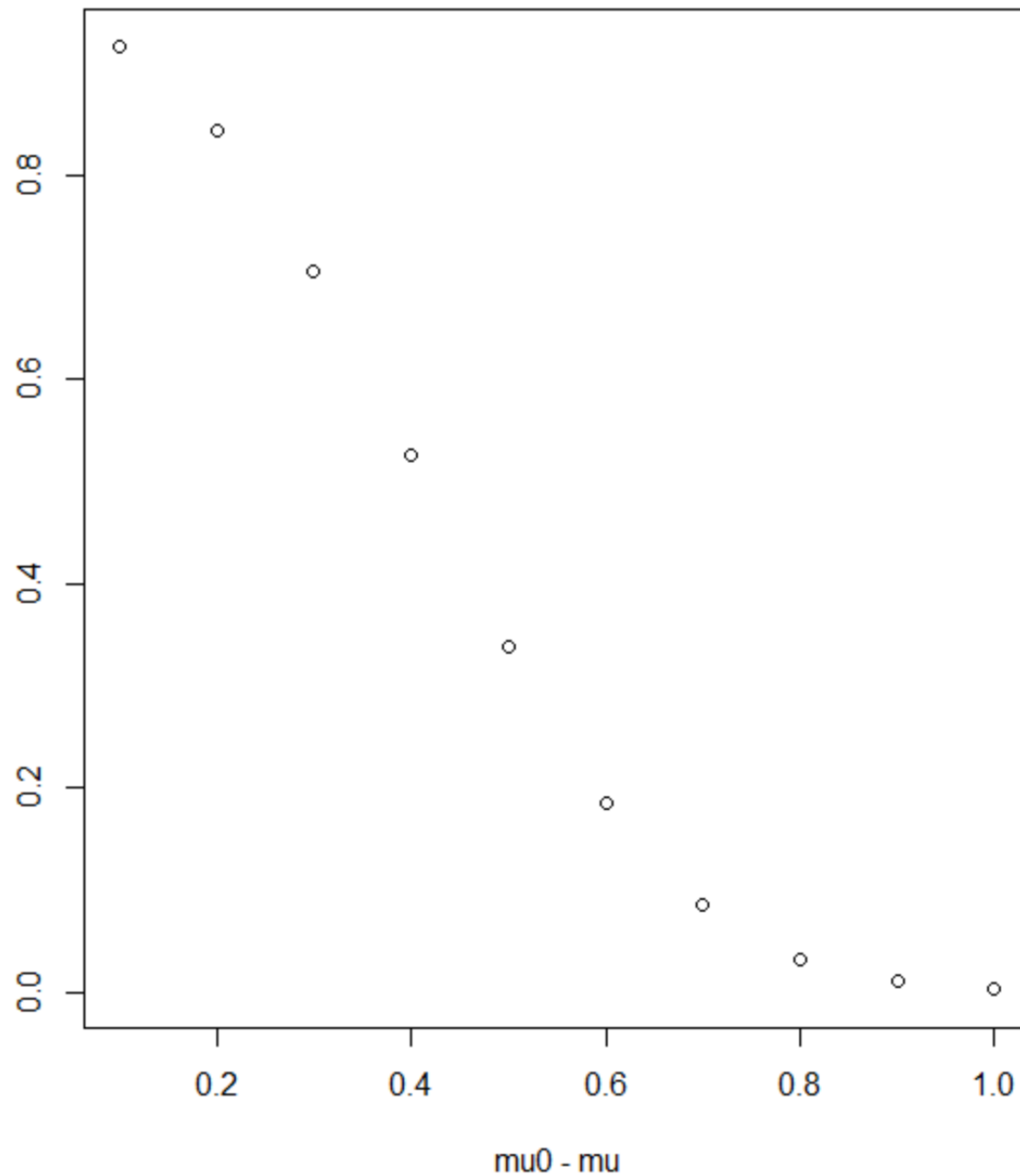
$$\begin{aligned} OC(\mu) &= P_{\mu}(-t_{n-1;1-\frac{\alpha}{2}} \leq \frac{\sqrt{n}(\bar{X}-\mu_0)}{S} \leq t_{n-1;1-\frac{\alpha}{2}}) \\ &= P_{\mu}(-t_{n-1;1-\frac{\alpha}{2}} \leq \frac{\sqrt{n}(\bar{X}-\mu)}{S} + \frac{\sqrt{n}(\mu-\mu_0)}{S} \leq t_{n-1;1-\frac{\alpha}{2}}) \\ &= P_{\mu}(-t_{n-1;1-\frac{\alpha}{2}} - \frac{\sqrt{n}(\mu-\mu_0)}{S} \leq \frac{\sqrt{n}(\bar{X}-\mu)}{S} \leq t_{n-1;1-\frac{\alpha}{2}} - \frac{\sqrt{n}(\mu-\mu_0)}{S}) \\ &= G_{n-1}\left(t_{n-1;1-\frac{\alpha}{2}} - \frac{\sqrt{n}(\mu-\mu_0)}{S}\right) - G_{n-1}\left(-t_{n-1;1-\frac{\alpha}{2}} - \frac{\sqrt{n}(\mu-\mu_0)}{S}\right). \end{aligned}$$

G_{n-1} is the distribution function for t_{n-1} (see page 4).

[If a random variable $X \sim F(x)$, then $P(a \leq X \leq b) = F(b) - F(a)$.]

One sample t -test – the OC function

```
mu=2 #the true value of the parameter  
mu0=seq(2.1,by=0.1,3)  
sample size = 100
```



```
mu=2 #the true value of the parameter  
mu0=2.1  
sample size = 200,300, ...,1000
```

